

Total-symbol Euler force, low-injected baseline transport, and block-Loewner boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-125 identifies the exact fixed-root total symbol

$$\mathcal{T}_k = \mathcal{V}_{\text{fut},k} - \mathcal{F}_{063,k}^{\text{ad}} \quad (1.1)$$

after all same-root visits have been recombined and the covariance owner $\Delta\Gamma_k$ is used once. It does not display the directional force of that symbol. It also leaves two apparently separate tasks: estimate the controlled secant and bound the root-only stationary baseline.

This note performs four proof-first audits of that boundary.

1. It differentiates (1.1) as one complete signed endpoint and gives its periodic Euler force. The derivative contains the covariance trace, current variation, derivative injection, and the surviving Cartan skew exactly once.
2. It distinguishes this Euler force from the predictable Cameron–Martin Riesz representative actually available to a legal control.
3. It transports the R-079 low, injected, and future blocks with every predictable-mean reserve and the generally nonzero finite-low reference $\mathbb{E}\mathcal{W}_J(A^0)$ retained.
4. It replaces a scalar off-diagonal norm slogan by the exact block-Loewner acceptance condition and computes the forward mixed-shell sums. It also records why neither the strong quadratic gap nor an unrestricted reverse extension is available.

The result is a finite-cutoff structural theorem, not the production one-use bound. It does not prove a forward, reverse, balanced, or low multiplier estimate for the actual projected force. It does not prove the production common-terminal identification used by the low/root diagnostic. It proves no $\text{OVERLAP}_{\text{src}}$, Nelson estimate, cutoff or floor removal, interacting measure, Sector-A closure, or tier promotion. The host claim remains T4.

Throughout Sections 2–5 the cutoff and positive floor are fixed. The heat law, output projection, covariance owner, and conditioning are fixed while an endpoint is varied. Coefficients are smooth bounded cylindrical objects, the covariance matrices are deterministic symmetric positive semidefinite owners, and all spatial integrations are over the torus. Complex rows require adjoints and real parts; the displayed proof uses the repository’s six-real coordinate.

2. The once-owned total trace-excess symbol

Fix a probability root k and suppress its already recombined visit index. Let

$$B(W) = C_6(W)^T C_6(W) = \sum_{a=1}^6 c_a(W) c_a(W)^T, \quad (2.1)$$

where c_a denotes one real row vector. For the spatial directions put

$$V_i = \partial_i W, \quad \Phi_{a,i}(W) = \mathbb{E}_f[c_a(W)^T V_i], \quad \bar{c}_a(W) = \mathbb{E}_f c_a(W). \quad (2.2)$$

Here $\mathbb{E}_f$ is exactly the R-125 future/heat conditioning: it may retain the current root. Define

$$\boxed{\mathcal{T}_k(W) = \sum_i \mathbb{E}_f \int B(W) : \Delta \Gamma_{k,i} \, dx - \sum_{a,i} \int |\Phi_{a,i}(W)|^2 \, dx.} \quad (2.3)$$

Under the R-125 exhaustive orthogonal cluster resolution, retained heat coordinate, and once-owned covariance hypotheses, (2.3) is the cluster sum of

$$S_{k,C}(W) = \Theta_{k,C}(W) - \|\Phi_{k,C}(W)\|^2. \quad (2.4)$$

The R-125 variance/forest bridge then gives

$$\mathcal{T}_k(W) = \mathcal{V}_{\text{fut},k}(W) - \mathcal{F}_{063,k}^{\text{ad}}(W). \quad (2.5)$$

If the output resolution is not exhaustive, the explicit R-125 complement leakage must be added. Contraction closure alone does not imply (2.5).

Theorem 2.1 (exact directional derivative)

Let $W_t = W + tH$ and $V_{i,t} = V_i + t\partial_i H$. The direction H may still belong to the retained finite-dimensional law; no force factorization is claimed at this stage. Then

$$\boxed{\begin{aligned} D\mathcal{T}_k(W)[H] &= \sum_i \mathbb{E}_f \int DB(W)[H] : \Delta \Gamma_{k,i} \, dx \\ &\quad - 2 \sum_{a,i} \text{Re} \int \Phi_{a,i}(W) \mathbb{E}_f [(Dc_a(W)H)^T V_i + c_a(W)^T \partial_i H] \, dx. \end{aligned}} \quad (2.6)$$

Proof. The covariance owner is fixed, so only $B(W)$ varies in the first term of (2.3). The product rule gives

$$\frac{d}{dt} \Phi_{a,i}(W_t)|_{t=0} = \mathbb{E}_f [(Dc_a(W)H)^T V_i + c_a(W)^T \partial_i H]. \quad (2.7)$$

Differentiate the Hilbert norm square and take its real polarization. This gives (2.6). No forest atom, coefficient derivative, or covariance trace has been estimated separately. $\square$

For a legal moving visit $W_{k,v,t} = W_{k,v-1} + ta_{k,v}$, summing (2.6) gives the exact endpoint telescope

$$\mathcal{T}_k(W_{k,V}) - \mathcal{T}_k(W_{k,0}) = \sum_v \int_0^1 D\mathcal{T}_k(W_{k,v,t})[a_{k,v}] \, dt. \quad (2.8)$$

A terminal straight homotopy is equally valid. A visit-square majorant is not invariant: two visits $a, -a$ have zero terminal displacement but positive sum of visit squares. Thus (2.8), not an arbitrary visitwise square, is the quotient-safe coordinate.

3. Periodic Euler force and the causal Riesz correction

Assume now that H is measurable with respect to the sigma-field retained outside $\mathbb{E}_f$, so it can be pulled through that conditional expectation. Using $DB[H] = \sum_a \{(Dc_a H)c_a^T + c_a(Dc_a H)^T\}$ and periodic integration by parts in the derivative-injection term yields the following.

Theorem 3.1 (Euler-force factorization)

Under the hypotheses above,

$$DT_k(W)[H] = \int H \cdot G_k(W) dx, \quad (3.1)$$

where

$$\begin{aligned} G_k &= 2 \sum_{a,i} \left\{ \mathbb{E}_f[(Dc_a)^T \Delta \Gamma_{k,i} c_a] + \partial_i(\Phi_{a,i} \bar{c}_a) - \Phi_{a,i} \mathbb{E}_f[(Dc_a)^T V_i] \right\} \\ &= 2 \sum_{a,i} \left\{ \mathbb{E}_f[(Dc_a)^T \Delta \Gamma_{k,i} c_a] + \bar{c}_a \partial_i \Phi_{a,i} + \Phi_{a,i} \mathbb{E}_f[(Dc_a - Dc_a^T) V_i] \right\}. \end{aligned} \quad (3.2-3.3)$$

Proof. The trace variation is

$$DB[H] : \Delta \Gamma_{k,i} = 2 \sum_a H^T (Dc_a)^T \Delta \Gamma_{k,i} c_a. \quad (3.4)$$

The current-coefficient variation gives the last term of (3.2). Periodic integration by parts sends

$$-2\Phi_{a,i} \bar{c}_a^T \partial_i H \quad \text{to} \quad 2H^T \partial_i(\Phi_{a,i} \bar{c}_a). \quad (3.5)$$

Finally $\partial_i \bar{c}_a = \mathbb{E}_f[Dc_a V_i]$, which converts the first line of (3.2) into its second line. $\square$

3.1 Curl convention

For the coefficient one-form $\vartheta_a = c_a(W)^T dW$, the repository exterior-curvature convention is

$$\Omega_a(H, V) = H^T (Dc_a^T - Dc_a) V. \quad (3.6)$$

The last term in (3.3) is therefore

$$-2\Phi_{a,i} \mathbb{E}_f[\Omega_a(\cdot, V_i)], \quad (3.7)$$

not $+2\Phi\Omega$. Calling $Dc_a - Dc_a^T$ itself the registered curl would reverse the sign of the R-121 2680/729 Cartan audit.

For the three linear rows, $Dc_a = 2\sqrt{c_0} S_A$ is symmetric and (3.7) vanishes. For the three rational rows, R-120 writes

$$Dc_a = 2\sqrt{c_1} F_A, \quad F_A = T_A - \alpha W \otimes g_A, \quad \alpha = \frac{5}{9}, \quad (3.8)$$

so the skew part survives. R-121's normalized owner audit is

$$-\frac{40}{729} + \frac{2720}{729} = \frac{2680}{729}. \quad (3.9)$$

Equation (3.3) retains this owner; it does not cancel it. The zeroth-order rational part still uses the R-120 fixed 21-matrix basis with its random scalar multipliers and spatial derivatives through order two.

3.2 Euler force is not yet an admissible Riesz vector

The object G_k is a distributional Euler force in the unweighted spatial duality. In a standard inhomogeneous H^2 metric its formal Riesz representative is

$$R_k^{H^2} = (1 - \Delta)^{-2} G_k. \quad (3.10)$$

For the actual quotient Cameron–Martin metric, the covariance/source map and its range projection must be used instead.

There is a second causal correction. Because $\mathbb{E}_f$ can retain root k , G_k is generally $\mathcal{F}_k$ -measurable, whereas a legal shell- k source is $\mathcal{F}_{k-1}$ -measurable. After the outer expectation,

$$\mathbb{E}\langle H_k, G_k \rangle = \mathbb{E}\langle H_k, P_{k-1} G_k \rangle. \quad (3.11)$$

Thus the admissible candidate has the schematic form

$$R_{k,\text{adm}} = \mathcal{R}_{\text{CM},k} P_{k-1} G_k, \quad (3.12)$$

followed by the chart/source range projection. Choosing $H_k = -G_k$ without (3.11)–(3.12) is anticipative and illegal. Moreover, a negative-gradient direction would prove only existence of one descent direction, not the uniform bound required for every Boue–Dupuis control.

4. Root/shell coordinates and the two forward scales

To remove the index ambiguity, let $r \geq j_0$ denote the probability root and let m denote the spatial output shell. The forward future region is

$$m \geq r + C. \quad (4.1)$$

R-093 and R-094 distinguish two scales. The positive quadratic curvature atom can have the strong prototype 2^{r-4m} . The complete mixed secant has a linear prototype 2^{r-2m} . The exact Hermite fixture has

$$-2\gamma\epsilon t + 4\gamma\epsilon^2 t^2, \quad (4.2)$$

so a nonzero linear term prevents promotion of the complete mixed symbol to a quadratic scale without a new signed cancellation. R-125’s abstract 2^{j-4k} lemma remains correct in its own root/shell naming convention; it is a conditional theorem, not the production factorization.

Lemma 4.1 (forward mixed geometric sum)

Suppose, after the total symbol and all matrix channels have been assembled, one has the declared block estimate

$$\|A_{r,m}^{\text{mix}}\| \leq C_{\text{mix}} 2^{r-2m}, \quad m \geq r + C. \quad (4.3)$$

Then its scalar-kernel square sum is

$$\sum_{r \geq j_0} \sum_{m \geq r+C} 2^{2r-4m} = \frac{64}{45} 2^{-4C} 2^{-2j_0}. \quad (4.4)$$

Its Hilbert–Schmidt majorant is therefore

$$K_{\text{mix}} \leq \frac{8C_{\text{mix}}}{\sqrt{45}} 2^{-2C} 2^{-j_0}. \quad (4.5)$$

At $C = 1$, $K_{\text{mix}} \leq 2C_{\text{mix}} 2^{-j_0} / \sqrt{45}$.

For comparison, the positive prototype has the exact ℓ^1 scalar sum

$$\sum_{r \geq j_0} \sum_{m \geq r+C} 2^{r-4m} = \frac{128}{105} 2^{-4C} 2^{-3j_0}. \quad (4.6)$$

Proof. Sum first in m . The ratios are 2^{-4} for both inner sums and 2^{-2} , respectively 2^{-3} , for the outer sums. Their products give $64/45$ and $128/105$. $\square$

Equations (4.3)–(4.6) are acceptance tests. This note does not prove (4.3) for the production force. If far and mixed contributions hit the same matrix block, the safe combination is

$$\|A_{\text{far}} + A_{\text{mix}}\| \leq K_{\text{far}} + K_{\text{mix}}. \quad (4.7)$$

Quadrature $\sqrt{K_{\text{far}}^2 + K_{\text{mix}}^2}$ requires disjoint matrix support or orthogonal domain/range channels with zero cross terms.

5. Exact block-Loewner acceptance condition

Let $R = R^*$ and $S = S^*$ collect signed diagonal reserves that remain in the same two-channel block as a mixed operator A . The desired upper payment is

$$\langle x, Rx \rangle + \text{Re}\langle x, Ay \rangle + \langle y, Sy \rangle \leq 2\eta\|x\|^2 + 2\zeta\|y\|^2. \quad (5.1)$$

Theorem 5.1 (block-Loewner criterion)

Equation (5.1) holds for all x, y if and only if

$$\boxed{\begin{pmatrix} 2\eta\text{I} - R & -A/2 \\ -A^*/2 & 2\zeta\text{I} - S \end{pmatrix} \succeq 0.} \quad (5.2)$$

For $R = S = 0$, this is equivalent to

$$\boxed{\|A\| \leq 4\sqrt{\eta\zeta}.} \quad (5.3)$$

If the adverse form is $2\text{Re}\langle x, Ay \rangle$, the threshold is instead $2\sqrt{\eta\zeta}$.

Proof. Subtract the left side of (5.1) from the right side and write the resulting quadratic form on $x \oplus y$. This is exactly the matrix in (5.2). With $R = S = 0$, its Schur complement is nonnegative exactly when $A^*A \preceq 16\eta\zeta\text{I}$. $\square$

Weighted or singular reserves require the corresponding generalized Schur complement, including kernel/range compatibility. One may not replace them by scalar η, ζ before the production matrix has been assembled.

For the disjoint R-103 debts and the R-124 legal-row source allocation, the trace-scale scalar budget is

$$K_{\text{bud}} = 4\sqrt{\left(\frac{197}{440} - \frac{3}{125P}\right)\frac{3}{25}} = 0.92093233\dots, \quad P = 4 + 10^{-12}. \quad (5.4)$$

Combining (4.5) with (5.4) at $C = 1$ would be sufficient if

$$C_{\text{mix}}2^{-j_0} < \frac{\sqrt{45}}{2}K_{\text{bud}} = 3.08890\dots \quad (5.5)$$

Again, (5.5) is not asserted for production.

6. The unrestricted reverse-band boundary

Fix a spatial shell m_0 and consider the normalized family

$$a^{(J)} = 2^{-2m_0} \phi(\xi_J) e_{m_0}, \quad J \longrightarrow \infty, \quad (6.1)$$

where ϕ is smooth bounded and centered. Its model H^2 scale is

$$2^{4m_0} \|a^{(J)}\|_2^2 \asymp \|\phi\|_2^2, \quad (6.2)$$

whereas coefficient-blind reverse weights 2^{J-4m_0} and 2^{J-2m_0} diverge. Hence no source-cost-only theorem may extend the forward kernels to an unrestricted reverse band.

This is a method boundary with narrow scope. For $J > m_0$, (6.1) reads future information and is not a legal strict-past production control. It does not refute the legal filtration, and it does not evaluate the complete signed variance-minus-forest/Cartan symbol. It is therefore not a production divergence counterexample. It forbids only the coefficient-blind anticipative extension; the legal reverse contribution must be derived from the actual causal projection in (3.11)–(3.12).

7. The balanced band remains critical

In bounded bandwidth the deterministic prototype is

$$\int |z|^2 |\nabla z|^2 \leq C_{\text{GN}}^2 \|z\|_{H^2} \|z\|_6^3 = C_{\text{GN}}^2 \sqrt{XY}. \quad (7.1)$$

For $z_{A,N} = A(\cos Nx, \sin Nx)$,

$$X \asymp A^2 N^4, \text{quad} Y = A^6, \text{quad} \int |z|^2 |\nabla z|^2 \asymp A^4 N^2 = \sqrt{XY}. \quad (7.2)$$

Thus a bounded root/shell collar creates no scaling slack by itself. The R-121 curl 2680/729 also forbids replacing the complete balanced current by a scalar primitive. The production balanced theorem must keep the trace, future variance, forest, rational zero-order multipliers, Cartan block, and mean reserve in the same Loewner matrix before taking positive parts.

8. Low-injected stationary-baseline transport

The root-only baseline of R-124 is not automatically signed. Nevertheless, R-079 gives an exact way to transport it without introducing an independent symbol, provided the production root family is first identified with the R-079 common-terminal Doob family and the low/root trace partition matches. These are hypotheses here, not consequences of R-125.

Let $e \in \{0, j, *\}$ denote the common reference, middle, and terminal endpoints. For each root/output owner put

$$\Phi_{j,C}^e = b_{j,C}^e + Y_{j,C}^e, \quad b_{j,C}^e = P_{j-1} P_C \mathcal{J}_e, \quad Y_{j,C}^e = d_j P_C \mathcal{J}_e, \quad (8.1)$$

$$D_{j,C}^e = \Theta_{j,C}^e - \|Y_{j,C}^e\|^2, \quad S_{j,C}^e = D_{j,C}^e - \|b_{j,C}^e\|^2. \quad (8.2)$$

The low owner has $Y_{\ell,C}^e = P_\ell P_C \mathcal{J}_e$ and $b_{\ell,C}^e = 0$.

Theorem 8.1 (R-079 sign ledger)

Under the common-terminal, exhaustive-output, and matching once-owned trace hypotheses,

$$\sum_{\mathcal{C}} \mathbb{E}(D_{j,\mathcal{C}}^j - D_{j,\mathcal{C}}^0) = -2\mathfrak{F}_j, \quad (8.3)$$

$$\sum_{\mathcal{C}} \mathbb{E}(D_{j,\mathcal{C}}^* - D_{j,\mathcal{C}}^j) = -2\mathfrak{J}_j, \quad (8.4)$$

and

$$\sum_{\mathcal{C}} \mathbb{E}(D_{\ell,\mathcal{C}}^* - D_{\ell,\mathcal{C}}^0) = -2\mathfrak{L}_{\ell}. \quad (8.5)$$

Consequently

$$D_*^{\text{full}} - D_0^{\text{full}} = -2 \left\{ \mathfrak{L}_{\ell} + \sum_j (\mathfrak{F}_j + \mathfrak{J}_j) \right\}, \quad (8.6)$$

and, with $M_e = \sum_{j,\mathcal{C}} \mathbb{E}\|b_{j,\mathcal{C}}^e\|^2$,

$$S_*^{\text{full}} - S_0^{\text{full}} = -2 \left\{ \mathfrak{L}_{\ell} + \sum_j (\mathfrak{F}_j + \mathfrak{J}_j) \right\} - (M_* - M_0). \quad (8.7)$$

Proof. R-079 defines $f_j = d_j(\mathcal{J}_j - \mathcal{J}_0)$ and $i_j = d_j(\mathcal{J}_* - \mathcal{J}_j)$. Expanding the two Hilbert squares in its definitions of $\mathfrak{F}_j$ and $\mathfrak{J}_j$ gives (8.3) and (8.4), with the minus sign because D is trace minus energy. The low polarization gives (8.5). Summation gives (8.6); subtracting predictable mean squares gives (8.7). $\square$

The same hypotheses imply

$$D_e^{\text{full}} = \mathbb{E}\Theta_e^{\text{tot}} - \mathbb{E}\|\mathcal{J}_e\|^2 = -2\mathbb{E}\mathcal{W}_J(A^e). \quad (8.8)$$

Therefore

$$S_0^{\text{full}} = -2\mathbb{E}\mathcal{W}_J(A^0) - M_0. \quad (8.9)$$

The finite-low reference $A^0 = A_{<j_0}$ is generally controlled. R-120 proves $\mathbb{E}\mathcal{W}_J(0) = 0$ only in its raw uncontrolled stationary scope; it does not permit deleting $\mathbb{E}\mathcal{W}_J(A^0)$ from (8.9).

For the middle/root baseline define

$$B_{\text{mid}} = \sum_{j,\mathcal{C}} \mathbb{E}S_{j,\mathcal{C}}^j, \text{quad} B_{\ell}^0 = \sum_{\mathcal{C}} \mathbb{E}S_{\ell,\mathcal{C}}^0. \quad (8.10)$$

Equations (8.3) and (8.9) cancel the reference root predictable means and give the sharp transport identity

$$B_{\text{mid}} = -B_{\ell}^0 - 2\mathbb{E}\mathcal{W}_J(A^0) - 2 \sum_j \mathfrak{F}_j - \sum_{j,\mathcal{C}} \mathbb{E}\|b_{j,\mathcal{C}}^j\|^2. \quad (8.11)$$

Thus

$$B_{\text{mid}} \leq 2C_0 \quad (8.12)$$

is equivalent, under the stated bridge hypotheses, to

$$B_{\ell}^0 + 2\mathbb{E}\mathcal{W}_J(A^0) + 2 \sum_j \mathfrak{F}_j + \sum_{j,\mathcal{C}} \mathbb{E}\|b_{j,\mathcal{C}}^j\|^2 \geq -2C_0. \quad (8.13)$$

Equation (8.13) removes a separately named baseline symbol only by coupling it to the injected block, finite-low endpoint, and favorable middle mean reserve. It does not prove their sign. Estimating the recombined complete endpoint instead is algebraically valid, but the R-093 directed-chart theorem shows that a uniform estimate of that endpoint is the original Nelson burden, not a bypass.

Three low objects remain distinct:

1. B_ℓ^0 , the current/trace low atom in (8.10);
2. $\mathcal{L}_\ell$, the R-079 conditional low-current endpoint difference containing $P_\ell \mathcal{J}_*$;
3. the R-079 complete low safe endpoint, equation (4.6) of that authority.

No two may be renamed as each other.

9. Smallest noncyclic successor

The exact total symbol is now explicit enough to state one unified successor instead of alternating between coefficient-first moments and an independent baseline sign.

First form G_k in (3.3) with every six-row output, covariance owner, future variance, adapted R-063 forest, rational multiplier, and the 2680/729 Cartan owner present. Then apply the predictable projection and actual Cameron–Martin Riesz/source map in (3.11)–(3.12). Decompose only this admissible total force into

$$G^{\text{adm}} = G^{\text{fwd}} + G^{\text{rev}} + G^{\text{bal}} + G^{\text{low}}. \quad (9.1)$$

The required theorem must:

1. prove the forward and legal reverse square ledgers from the quotient source norm, with no visit or chart multiplicity;
2. keep the linear mixed scale separate from the positive quadratic atom;
3. assemble every same-block contribution before applying the Loewner test (5.2), using triangle rather than unproved quadrature;
4. prove the critical balanced aggregate with the finite-low and injected transport (8.13) retained;
5. be uniform in cutoff, floor, chart, control, dimension, output cluster, same-root subdivision, and representation-preserving refinement.

If this single theorem meets the remaining R-103/R-124 budget, R-093 can be invoked over the directed chart union. Neither the theorem nor its hypotheses are proved here.

10. Evidence and route decisions

The primary executable performs an exact finite-atom differentiation of (2.3), an exact trigonometric torus check of both forms in (3.2)–(3.3), the R-079 sign ledger including nonzero $\mathcal{W}_J(A^0)$, the block-Loewner boundary, the two geometric sums, the production budget interval, the R-121 owner curl, the linear mixed-secant obstruction, and the unrestricted reverse-band scale. The independent standard-library executable uses polynomial arithmetic, periodic quadrature exact for the chosen trigonometric fixture, a different low/root rational fixture, and separate scalar-matrix checks.

The reusable route verdicts are:

1. **Advanced:** the R-125 total symbol has the exact derivative (2.6) and predictable Euler coordinate (3.3).
2. **Corrected:** G_k is not yet the legal Riesz vector; predictability and the actual CM/source map are load-bearing.
3. **Boundary:** the full mixed secant has a linear scale; the strong quadratic gap belongs only to the declared positive atom unless a new cancellation is proved.

4. **Advanced conditionally:** (4.4)–(5.5) give exact acceptance constants after a production block estimate exists.
5. **Failed method:** a coefficient-blind forward kernel cannot be extended to an unrestricted anticipative reverse band. This is not a legal production-control counterexample.
6. **Advanced conditionally:** (8.11)–(8.13) merge the root baseline with low and injected owners, but only under the missing production common-terminal/matching-trace identification and with no sign gained.
7. **Open:** the unified admissible-force estimate in Section 9.

11. Devil’s-advocate review

1. **Objection: the derivative (2.6) already proves a favorable sign. UPHELD AGAINST THE CLAIM.** A directional derivative has no sign or convexity. The required result is a uniform bound for every legal control.
2. **Objection: choose the control to be minus the force. UPHELD AGAINST THE CLAIM.** The raw force can be $\mathcal{F}_k$ -measurable; the legal direction is $\mathcal{F}_{k-1}$ -measurable. Equations (3.11)–(3.12) and the source range are mandatory.
3. **Objection: (3.3) cancels the Cartan block. UPHELD AGAINST THE CLAIM.** The row curl convention has the opposite matrix order. R-121’s nonzero 2680/729 owner survives.
4. **Objection: every far term enjoys 2^{r-4m} . UPHELD AGAINST THE CLAIM.** The R-094 mixed secant has the nonzero linear term (4.2). Only the positive quadratic atom has the stronger prototype without a new cancellation.
5. **Objection: far and mixed Hilbert–Schmidt constants may always be added in quadrature. UPHELD AGAINST THE CLAIM.** Shared matrix blocks have cross terms. Equation (4.7) is the safe rule unless orthogonality is proved.
6. **Objection: the reverse fixture disproves the production filtration. DISMISSED.** The fixture is anticipative. It refutes only an unrestricted coefficient-blind reverse extension.
7. **Objection: low-retained recombination proves $C_0 = 0$. UPHELD AGAINST THE CLAIM.** Equation (8.13) still contains the finite-low endpoint, injected block, and the missing common-terminal identification. It is an equivalence, not a sign theorem.
8. **Objection: $\mathbb{E}\mathcal{W}_J(A^0) = 0$ by stationarity. UPHELD AGAINST THE CLAIM.** R-120 proves the statement only for the raw uncontrolled endpoint. R-079’s A^0 retains finite-low control.
9. **Objection: the abstract Loewner threshold is the production constant. UPHELD AGAINST THE CLAIM.** The matrix A , its diagonal reserves, kernel compatibility, and its uniform norm remain to be constructed from the complete production symbol.

12. Result footer

Result ID: A13-CLASSII-TOTAL-SYMBOL-EULER-LOW-INJECTED-
LOEWNER-BOUNDARY (R-126).

Precise statement: Theorems 2.1, 3.1, 5.1, and 8.1; Lemma 4.1;
unrestricted reverse-band and balanced-band boundaries in Sections 6–7.

Scope: fixed finite cutoff and positive floor; fixed common legal heat,

conditioning, output projection, and symmetric once-owned covariance;
exhaustive orthogonal cluster resolution for identification with R-125;
smooth bounded cylindrical coefficients; periodic torus; predictable
direction for the Euler-force factorization; common-terminal/matching-trace
hypotheses only for Section 8.

Dependencies: A1; R-063, R-074, R-079, R-093, R-094, R-103,
R-104, R-118, R-120, R-121, R-123, R-124, R-125.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_total_symbol_euler_low_injected_
loewner_boundary_verify.py

Expected output: R-126 integrated PASS; primary and independent children,
note/PDF, manifest, exploration, negative, and public-surface gates pass.

Falsification gate: any failed total-symbol derivative, factor of two,
transpose/curl sign, causal projection scope, R-079 sign or W(A0) owner,
geometric factor, Loewner threshold, reverse-fixture scope, authority/hash,
PDF, or public-surface check.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no production admissible-force operator estimate,
forward/reverse/balanced/low closure, common-terminal production theorem,
stationary-baseline sign, OVERLAP_src, Nelson, removals, interacting
measure, Sector-A closure, or T5-T7 follows.

Next required action: construct the predictable CM/source-projected total
force, prove one unified forward/reverse/balanced/low block-Loewner estimate
with the low/injected transport retained, then invoke the R-093 directed
chart union only if the production budget is met.